

SEATTLE TACOMA, WA, USA

WMO: 727930

Lat: 47.444N

Lon: 122.314W

Elev: 370

StdP: 14.50

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 24233

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	26.6	30.0	10.5	9.5	32.8	16.6	12.9	34.5	26.5	46.2	23.9	47.1	8.1	10	0.531

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.0	86.1	65.1	82.2	63.9	78.7	62.7	66.7	82.7	65.1	79.5	63.6	76.5	8.9	350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.0	81.1	69.5	59.5	76.9	68.5	58.3	73.6	67.5	31.3	82.9	30.1	79.2	29.0	76.4	83.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.4	18.1	16.2	DB	21.8	92.7	4.2	4.2	18.8	95.7	16.3	98.1	13.9	100.4	10.8	103.4	(4)
(5)			WB	19.6	70.1	4.4	3.2	16.4	72.4	13.9	74.3	11.4	76.1	8.2	78.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	53.0	42.3	43.1	46.3	50.4	56.5	61.2	66.5	66.6	61.8	52.9	46.0	41.3
	DBStd	10.38	5.79	5.05	4.91	5.00	5.58	5.40	5.17	4.48	4.65	4.38	5.54	5.53
	HDD50	1067	243	196	135	52	7	0	0	0	0	21	141	272
	HDD65	4621	704	613	579	439	272	142	40	33	119	375	571	734
	CDD50	2147	4	3	22	63	207	335	512	514	353	111	20	3
	CDD65	226	0	0	0	1	7	27	87	82	22	0	0	0
	CDH74	2047	0	0	2	15	97	283	815	657	177	1	0	0
	CDH80	582	0	0	0	2	18	86	266	178	32	0	0	0
Wind	WSAvg	7.6	8.1	8.4	8.5	7.8	7.4	7.4	7.0	6.6	6.7	7.1	8.0	8.4
Precipitation	PrecAvg	38.3	5.6	3.9	3.9	2.8	1.8	1.5	0.7	1.1	1.7	3.6	6.2	5.7
	PrecMax	51.8	11.7	9.1	9.4	6.5	3.7	3.8	2.4	4.6	6.2	10.0	15.6	11.9
	PrecMin	25.1	0.6	0.4	0.6	0.6	0.1	0.2	0.0	0.0	0.0	0.3	0.7	1.4
	PrecStd	6.3	2.2	2.0	1.8	1.4	0.9	0.9	0.5	1.1	1.4	2.1	2.8	2.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.1	58.6	67.9	75.9	82.4	88.2	92.5	89.7	84.1	70.2	61.8	56.6
		MCWB	50.8	48.0	52.5	57.6	61.6	64.6	67.7	66.3	62.8	57.3	55.0	52.1
	2%	DB	54.0	55.2	61.2	68.0	76.2	81.9	86.8	84.7	79.2	65.5	57.9	53.1
		MCWB	49.0	48.2	50.6	54.2	59.9	62.9	66.0	65.2	62.5	55.9	53.4	48.9
	5%	DB	51.8	52.6	57.1	63.0	71.4	76.3	82.2	81.1	74.8	62.4	55.5	50.7
		MCWB	47.6	47.0	49.2	52.1	57.6	60.6	64.4	63.9	61.1	55.2	51.7	46.9
	10%	DB	50.0	50.3	54.0	59.1	66.8	71.7	78.2	77.3	71.0	59.7	53.4	48.8
		MCWB	46.2	45.5	47.8	50.1	55.9	59.0	62.9	62.9	59.6	54.1	49.7	45.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	54.2	53.0	54.9	59.3	63.5	67.2	70.2	68.1	66.2	60.9	57.5	53.6
		MCDB	55.5	54.7	61.8	72.7	76.0	84.9	88.9	84.6	78.9	65.0	60.0	55.9
	2%	WB	50.9	50.3	52.4	55.4	61.2	63.9	67.2	66.3	63.9	58.4	54.5	50.2
		MCDB	52.5	53.0	57.8	64.5	73.4	78.4	83.7	81.8	75.3	62.7	56.9	52.2
	5%	WB	48.6	48.2	50.7	53.2	59.0	61.6	65.2	64.9	62.2	56.5	52.6	47.9
		MCDB	50.5	51.1	55.7	60.6	69.8	73.9	80.2	78.9	72.2	60.4	54.9	50.2
	10%	WB	46.7	46.4	48.6	51.2	56.8	59.7	63.5	63.5	60.6	54.9	50.6	45.9
		MCDB	49.1	49.5	53.4	57.3	65.5	70.2	76.7	76.0	69.0	58.6	53.0	48.5
Mean Daily Temperature Range	5% DB	MDBR	8.8	10.7	12.4	14.2	15.6	16.2	19.0	18.5	15.9	11.6	9.7	8.3
		MCDBR	10.5	13.7	17.4	21.6	24.3	24.3	26.1	25.0	22.9	16.4	11.3	9.3
		MCWBR	7.4	8.5	9.4	10.6	11.1	10.2	10.1	9.6	9.5	8.8	7.7	6.9
	5% WB	MCDBR	8.7	10.5	14.7	18.4	21.3	21.7	24.3	22.8	20.1	13.3	9.7	8.4
		MCWBR	7.3	7.8	8.9	9.6	10.5	9.9	10.0	9.5	9.0	8.2	7.6	7.4
Clear-Sky Solar Irradiance	taub	0.324	0.315	0.323	0.355	0.373	0.363	0.349	0.355	0.343	0.327	0.319	0.329	
	taud	2.494	2.548	2.546	2.426	2.396	2.447	2.507	2.497	2.521	2.571	2.542	2.409	
	Ebn at Noon	241	273	287	285	283	285	288	282	276	264	242	220	
	Edh at Noon	21	23	27	34	36	35	32	31	28	23	19	20	
All-Sky Solar Radiation	RadAvg	324	601	903	1326	1669	1835	2052	1750	1274	723	379	264	
	RadStd	46	80	103	123	136	179	151	116	114	79	50	38	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	+1.84	N/A	N/A	N/A	N/A	+223	+77
Regional (57 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page